

4. EAX
EBX
ECX
EDX

ESP
ESI
EBP
EDI

Assembly Language

Homework 3

Deadline: Monday October 27th, 2025 23:59

1. The central processor unit (CPU) contains registers and what other basic elements? *ALU, CU, Clock*
2. The central processor unit is connected to the rest of the computer system using what three buses? *data bus, control bus, address bus*
3. What are the three basic steps in the instruction execution cycle? *3. assembler*
4. Name all eight 32-bit general-purpose registers. *5. CS ES DS FS SS GS linker OS loader*
5. Name all six segment registers.
6. Name at least four CPU status flags. *6. CF ZF OF SF*
7. Which flag is set when the result of an unsigned arithmetic operation is too large to fit into the destination? *CF*
8. Which flag is set when the result of a signed arithmetic operation is either too large or too small to fit into the destination? *OF*
9. Explain the difference between big endian and little endian.
10. Why might you use a symbolic constant rather than an integer literal in your code?
11. Given the number 456789ABh, list out its byte values in little-endian order.
12. Declare an array of 120 uninitialized unsigned doubleword values.
13. Declare a 32-bit signed integer variable and initialize it with the smallest possible negative decimal value. (Hint: Refer to integer ranges in Chapter 1.)
14. Declare an uninitialized array of 50 signed doublewords named dArray.
15. Show the order of individual bytes in memory (lowest to highest) for the following doubleword variable:

val1 DWORD 87654321h

Val1 DWORD 12345678h

9. big endian

little endian

10. Symbolic constants make me easier to read in code and also they are easy to maintain.

11. 456789ABh

15.

0000	12
0001	34
0002	56
0003	78

0000	78
0001	56
0002	34
0003	12

AB
89
67
45

21
43
65
87

12.

array DWORD 120 DUP(?)

13.

Var SDWORD -2147483648

14.

dArray SDWORD 50 DUP(?)